A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Renewable energy technologies - Set 1

Mark scheme

AQA spec 3.3.4.1; Energy resources; textbook pp. 196-217

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged **1** Electrons flow through an external conductor / circuit to the relatively positive lower layer – the current powers appliances **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Smooth cells reflect about 30% of the light – grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away **1** Multi-junction cells have several layers of different materials, each absorbing different wavelengths **1** So a greater proportion of the available light is absorbed and converted to electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Solar farm:** polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m^2 matters less) **1 Satellite:** multi-junction - highest efficiency where mass / area is critical and cost is less important **1** Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1 Intermittent:** no output at night and low in winter (daily and seasonal cycles) - predictable **1 Unpredictable:** cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** A fluid in a buried ground loop absorbs low-temperature heat from the ground – at the expansion valve the fluid boils at low pressure absorbing this heat **1** A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building **1** The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures **1** The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C **1** The stored heat boils water to drive steam turbines whenever electricity is required, including at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight – cloud, dust and smoke scatter light so it cannot be concentrated **1** PV cells convert diffuse as well as direct light, though output is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **diffuse** → pollution from a wide area with no single outlet	2
02.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1 Manufacture:** extraction / processing of silicon, metals and plastics – toxic wastes (silicon tetrachloride, cadmium) **1** Land take by large solar farms competing with other uses / affecting desert flora and fauna **1** Water for cleaning panels in arid areas **1** Disposal of old panels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 5 points below. **1** No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) **1** Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas **1** Small scale matches local demand – electricity used locally without long transmission lines / grid connection **1** Low construction cost and simple technology maintained by the community **1** Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed – Kaplan turbines with adjustable blades cope with variable flow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	5
03.2	1 mark for the point below. **1** Lack of capital / finance, technical knowledge or grid access – seasonal drying of rivers – conflicts with irrigation water – planning / land ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Output is proportional to flow x efficiency: 100 x 0.10 x (40 / 90) **1** = 4.4 kW (accept 4.4-4.5) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (kW) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost – high flows that clear gravel spawning beds lost – floodplain no longer flooded **1** Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis **1** Sediment trapped so less nutrient replenishment downstream – barrier to migrating fish (salmon, sturgeon) and isolation of populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10^3 / 6^3 = 1000 / 216 **1** = 4.6 times (accept 4.63) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 5 points below. **1** C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs **1** A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release) – far from grid – scenic objections **1** B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting **1** D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy **1** Judgement, eg C or B – turbines spaced 3-5 blade diameters apart to avoid wind shadow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	5
04.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Driven by wind from any direction so no motor to turn them – suitable for turbulent urban sites **1** Generator can be at ground level so a lighter tower **1** Quieter **1** Operate at lower wind velocities **1** Blades more visible so fewer bird deaths **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
04.4	1 mark for the point below. **1 No gearbox:** fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** High mean wind velocities that are consistent in strength and direction **1** A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.2	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber – the air forced in and out drives a turbine – surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine) – overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines) – oscillating wave surge converter (Oyster) **OR** point absorber (floating buoy moving relative to a fixed base turns a generator) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
05.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Equipment must withstand storms and corrosion **1** Difficult to anchor off stormy coasts / in deep water **1** Expensive to transmit electricity from isolated coasts to consumers **1** Competition with established technologies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Barrage:** huge quantities of construction materials – reduces tidal range so intertidal mudflats used by wading birds are lost – changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity – pollutants concentrate in the lagoon – blocks migratory fish and shipping **up to 2 Lagoon:** similar but smaller scale – only part of the estuary enclosed so the most sensitive areas and main channel can be avoided **up to 2 In-stream turbines:** do not block the tidal flow so little impact on range or sediment – but noise affects cetaceans and many dispersed turbines could have a larger total impact – harness less energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	
06.1	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines – no generation when levels are equal (slack water) **OR** possible on the flood when the sea is higher than the basin **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow – the timing does not match demand **1** Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1 Spring tides:** Sun and Moon aligned so their gravitational effects combine giving a large tidal range – neap: effects oppose giving a small range **1** Energy depends on the mass of water and height difference, so output is much greater at springs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** When demand is low, surplus electricity pumps water into the lagoon above sea level **1** The stored water is released through the turbines later when demand is high (pumped storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak	2
06.5	**1 mark each** 1 mark for each of the 2 points below. **1** Large tidal range (eg Severn Estuary) **1** Estuaries, bays or headlands / channels that focus the flow and increase velocity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) **1** Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles **1 Carbon neutral in principle:** CO2 released equals CO2 absorbed in photosynthesis **1** Energy density of alcohol / biodiesel nearly as high as fossil fuels – wastes (straw, sewage, landfill gas) are cheap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **landfill gas** → the methane and carbon dioxide produced by anaerobic decomposition of waste **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
07.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) **1** Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO2 as using fossil fuels directly **1** Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally **1** Supply from wastes limited to the amount produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Hydrogen from algae:** some algae deprived of sulfur produce hydrogen during photosynthesis **1** Anaerobic digestion extended to crop wastes and marine algae – small-scale digesters (eg one cow's manure supplying a family's cooking gas) **1** Biofuels from microorganisms (eg oil-producing microalgae) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK **1** New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Properties of renewables:** intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • **Wind:** large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments – intermittent, needs storage and grid – land-use conflicts • **Solar:** PV costs falling rapidly, roof space, but low winter insolation in the UK – CSP unsuitable – heat pumps and photothermal for heating • **HEP:** most large sites used – low-head, helical and micro-hydro extend use – pumped storage vital • **Tidal:** large range (Severn, Mersey), predictable – barrages costly with high impact, lagoons and in-stream turbines lower impact • **Wave:** long Atlantic fetch, but technology immature • **Biofuels:** wastes, anaerobic digestion, energy crops vs food / habitat • **Geothermal:** limited to low-temperature schemes / heat pumps • **Systems issues:** storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • **Judgement:** renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder – transition needs investment and lifestyle change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any accurate, relevant material that is not in the indicative content. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **insolation** → incoming solar radiation / energy arriving from the Sun • **HEP** → electricity generated from falling water **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (8 marks)	
	What renewables can contribute depends on their properties - intermittency, predictability, energy density, storage, applicability, location, impacts, technological maturity and cost - set against the UK's resources. Wind is the largest: shallow seas give a huge offshore resource, costs have fallen and direct-drive and vertical-axis designs keep improving, but it is intermittent, needs storage and grid connections and provokes land-use conflicts onshore. Solar PV costs are falling rapidly and roof space is plentiful, but winter insolation is low and concentrating solar is unsuitable; photothermal panels and heat pumps are better for heating. Most large HEP sites are used, though low-head, helical and micro-hydro extend it and pumped storage is vital for balancing. The Severn and Mersey have large, predictable tidal ranges, but barrages are costly and damaging while lagoons and in-stream turbines have lower impact; the long Atlantic fetch gives wave energy that the technology cannot yet harness reliably. Biofuels from wastes and anaerobic digestion are useful, but energy crops compete with food and habitat, and geothermal is limited to low-temperature schemes.	
	The system matters as much as the sources: storage in pumped HEP, batteries, hydrogen and power-to-gas, vehicle-to-grid, grid management, a diverse mix that smooths supply, and conservation to reduce demand. With these, renewables could supply most of the UK's electricity. Heat and transport are harder, because they need high temperatures or liquid fuels, so electrification and hydrogen would be required. The potential is large, but realising it needs sustained investment and changes in how energy is used.	
	Weaker answer - Level 1 (3 marks)	
	The UK could use lots of renewable energy. It is windy so wind turbines could be built, especially offshore. Solar panels can be put on roofs but it is not very sunny in winter. Tidal power could be used in the Severn estuary because it has big tides. Hydroelectric power can be used in Scotland. Renewables are intermittent so batteries would be needed. Renewables could provide a lot of the UK's energy in the future.	
	What makes the difference	
	• Level 1 mentions the main resources with one fact each; Level 3 assesses each against its properties, constraints and the state of the technology.	
	• Level 3 treats storage, grid and demand as a system and distinguishes electricity from heat and transport - essential for judging 'future energy supplies'.	

Do not write outside the box

Question	Marking guidance	Marks
	• The conclusion in Level 3 is qualified (most electricity; heat and transport harder; needs investment); Level 1's is unqualified.	
08.2	1 mark for the point below. **1** Eight times (energy proportional to velocity cubed) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Do not write outside the box